

Recall from last class:

- Let $u, v \in \mathbb{R}^n$, then the **dot product** of u and v is given by

$$u \cdot v = u^T v = u_1 v_1 + \cdots + u_n v_n.$$

- **Theorem 6.1:** Let $u, v, w \in \mathbb{R}^n$ and $c \in \mathbb{R}$.

1.

$$u \cdot v = v \cdot u.$$

2.

$$(u + v) \cdot w = u \cdot w + v \cdot w.$$

3.

$$(cu) \cdot v = c(u \cdot v) = u \cdot (cv).$$

4. $u \cdot u \geq 0$ and $u \cdot u = 0$ if and only if $u = 0$.

- The **length** or **norm** of the vector v is a non-negative scalar, denoted by $\|v\|$, and defined by

$$\|v\| = \sqrt{v \cdot v}.$$

- We define the **distance** between u and v by

$$\text{dist}(u, v) = \|u - v\|.$$

- The vectors u and v are **orthogonal** if $u \cdot v = 0$.

- **Theorem 6.2:** The vectors u and v are orthogonal if and only if

$$\|u\|^2 + \|v\|^2 = \|u + v\|^2.$$

- The set

$$W^\perp = \{z \in \mathbb{R}^n : z \cdot w = 0 \text{ for all } w \in W\}$$

is called the **orthogonal complement** of W .

- Note: $x \in W^\perp$ if and only if x is orthogonal to each vector in a set that spans W . Also $W^\perp$ is a subspace of $\mathbb{R}^n$.

- We can use the dot product to find the angles between vectors; namely

$$u \cdot v = \|u\| \|v\| \cos \theta.$$

- A set of vectors $\{u_1, \dots, u_p\}$ in $\mathbb{R}^n$ is **orthogonal** if $u_k \cdot u_j = 0$ for all $j \neq k$.

- **Theorem 6.3** Let A be an $m \times n$ matrix, then

$$(\text{Row}(A))^\perp = \text{Null}(A) \text{ and } (\text{Col}(A))^\perp = \text{Null}(A^T).$$

- **Theorem 6.4:** If $S = \{u_1, \dots, u_p\}$ is an orthogonal set of non-zero vectors in $\mathbb{R}^n$, then S is linearly independent and hence a basis for $\text{Span } S$.

Definition 1. An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

Theorem 6.5: Let $\{u_1, \dots, u_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $y \in W$, then

$$y = c_1 u_1 + \cdots + c_p u_p$$

where

$$c_j = \frac{y \cdot u_j}{u_j \cdot u_j}.$$

1. Prove the Theorem.

2. Let $u_1 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and $u_2 = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$ and $x = \begin{bmatrix} 9 \\ -7 \end{bmatrix}$. Find x in the $\{u_1, u_2\}$ -coordinates.

Definition 2. Let u be a non-zero vector in $\mathbb{R}^n$ and let v be a vector in $\mathbb{R}^n$, then the **orthogonal projection** of v onto u , denoted by $\text{proj}_u v$, is given by

$$\text{proj}_u v = \frac{v \cdot u}{u \cdot u} u$$

Let $L = \text{Span}\{u\}$, then we also write $\text{proj}_L v$ for the orthogonal projection of v onto the line defined by the span of u ; namely

$$\text{proj}_L v = \text{proj}_u v.$$

3. Let $u = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$. Find the orthogonal projection of y onto u .

Note: Theorem 6.5 says that

$$y = \text{proj}_{u_1} y + \text{proj}_{u_2} y + \cdots + \text{proj}_{u_p} y.$$

Definition 3. A set $\{u_1, \dots, u_p\}$ is **orthonormal** if it is orthogonal and $\|u_k\| = 1$ for all k .

Theorem 6.6: An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I_n$.

4. Prove the Theorem.

Theorem 6.7: Let U be an $m \times n$ matrix with orthonormal columns. Let $x, y \in \mathbb{R}^n$. Then

- (a) $\|Ux\| = \|x\|$.
- (b) $(Ux) \cdot (Uy) = x \cdot y$.
- (c) $(Ux) \cdot (Uy) = 0$ if and only if $x \cdot y = 0$.

5. Prove the theorem.

Definition 4. An **orthogonal matrix** U is a square matrix so that $U^{-1} = U^T$.

Note: An $n \times n$ matrix U is orthogonal if and only if the columns of U are orthonormal.

6. Suppose U is an orthogonal matrix.

- (a) What can you say about the eigenvalues of U ?

- (b) What is the determinant of U ?